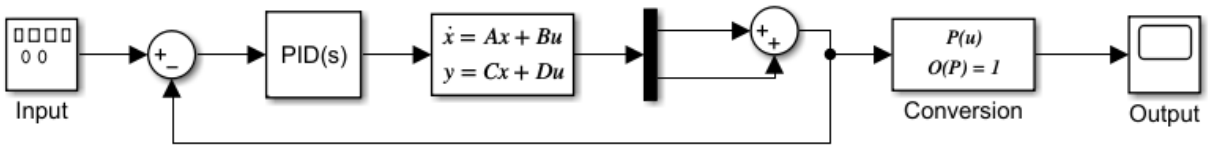
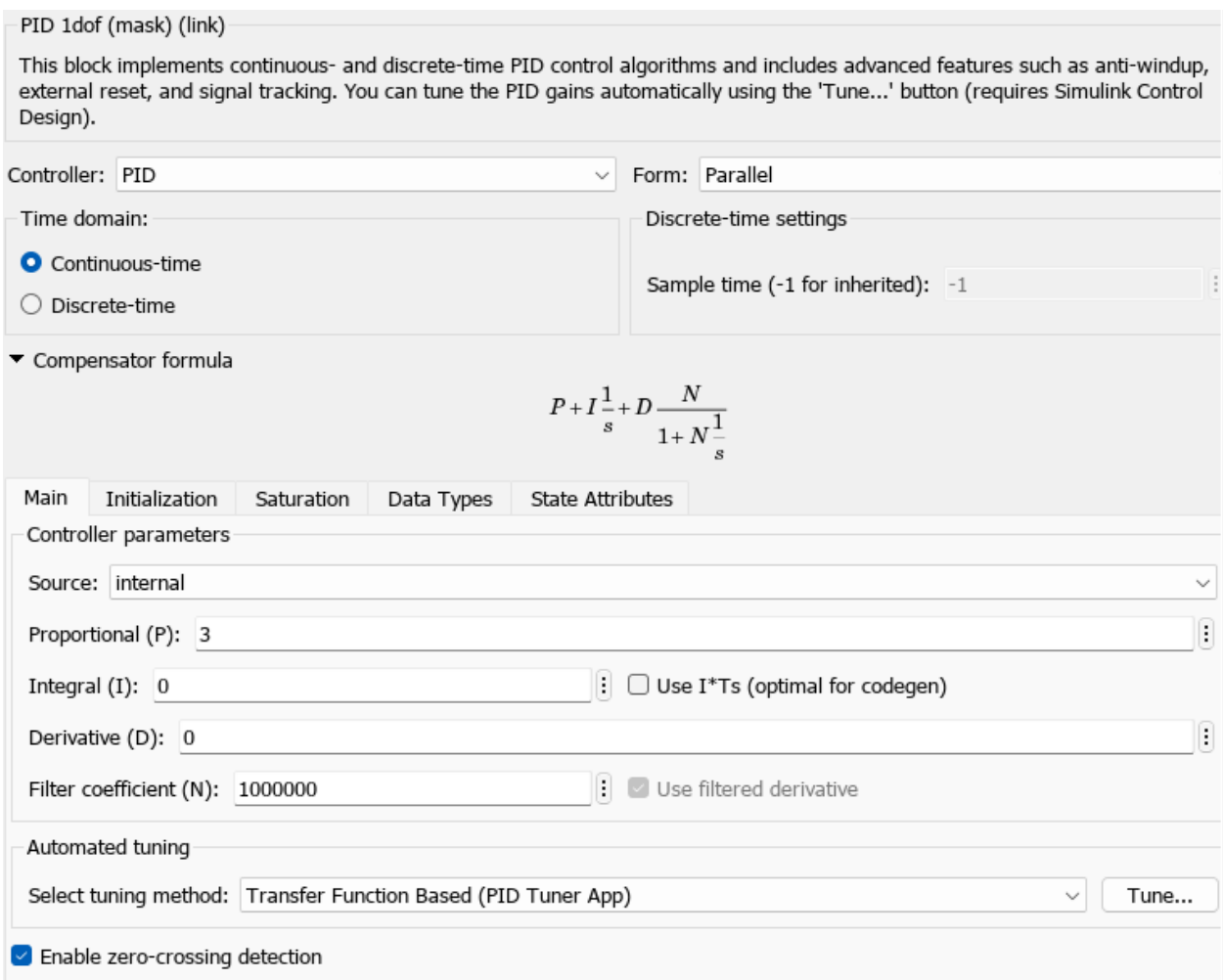
Haoming Ye 260896821

i)

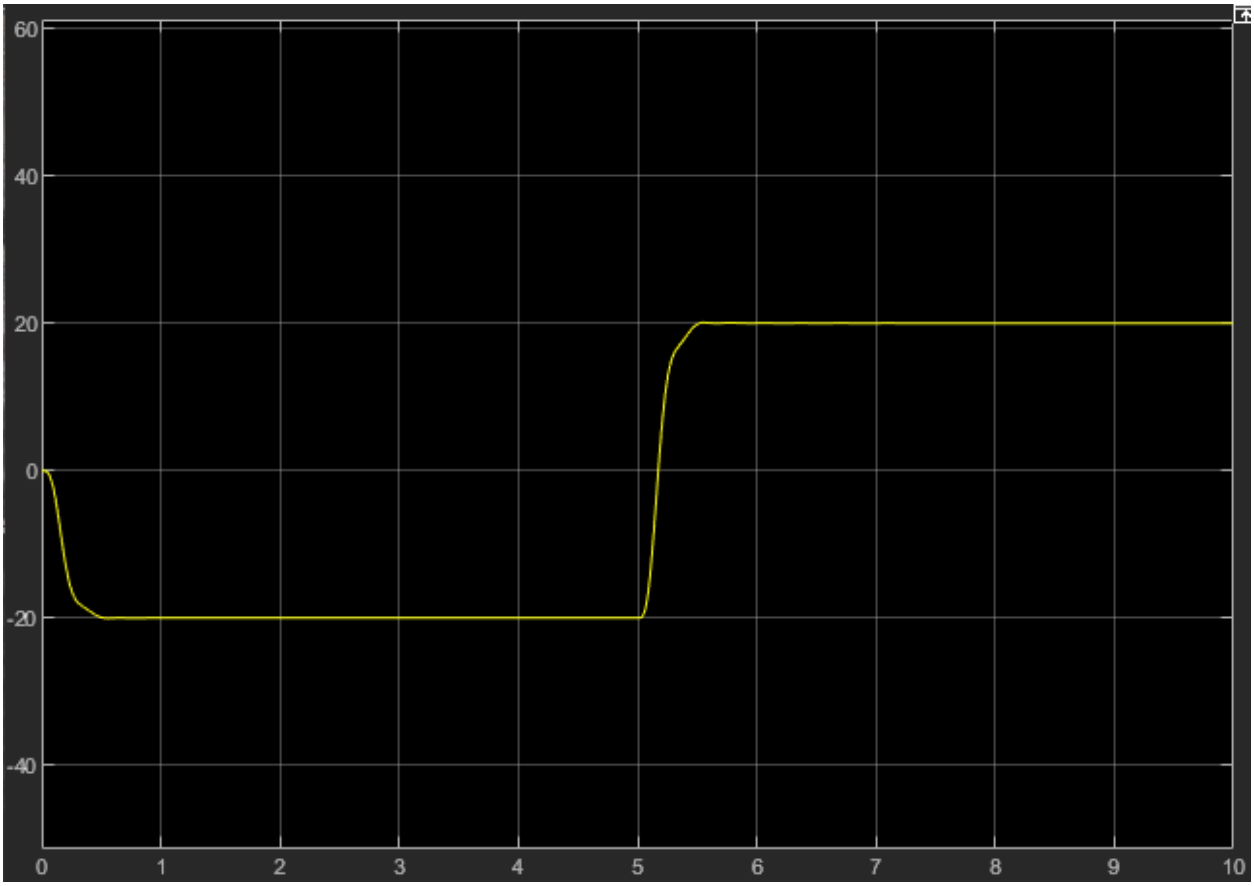


ii)

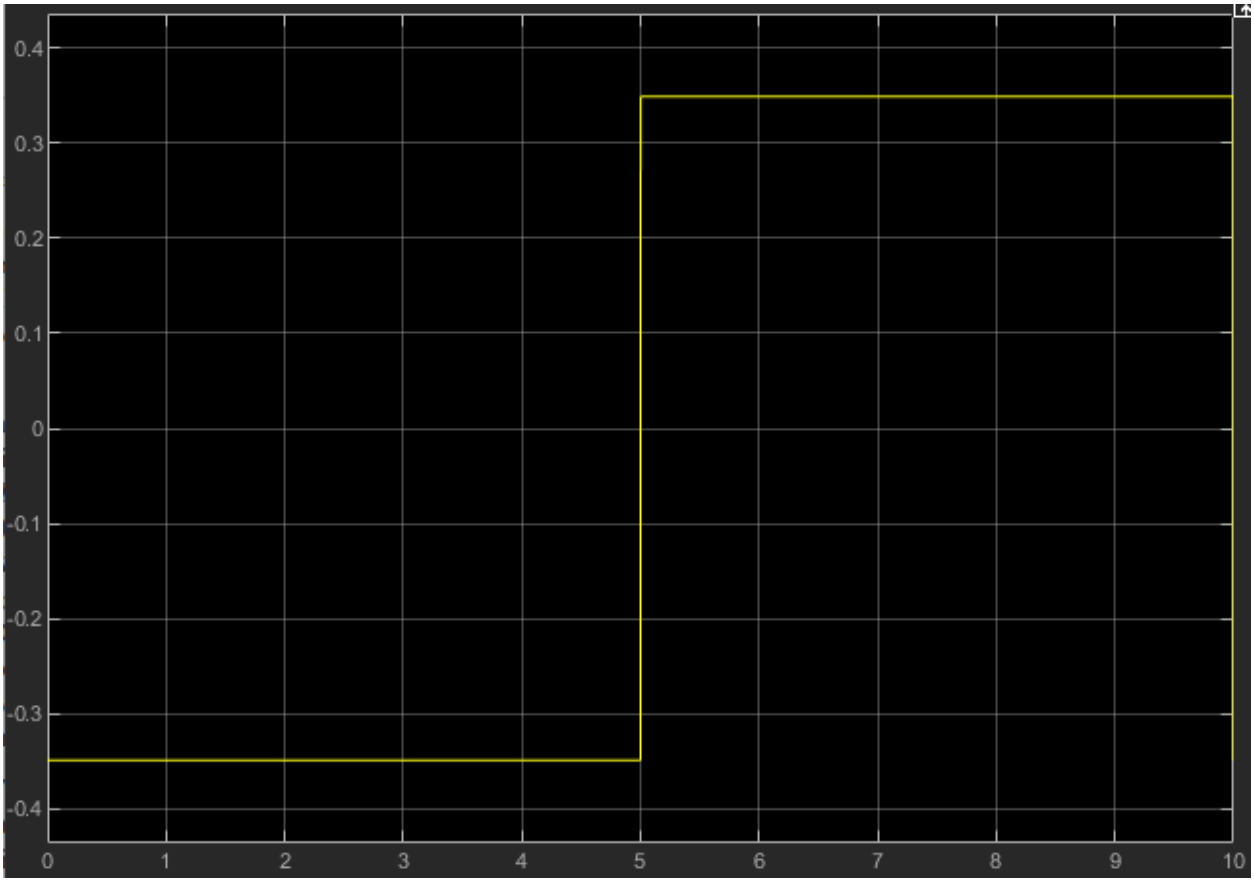
P:



Theta+alpha vs. time



Voltage vs. time

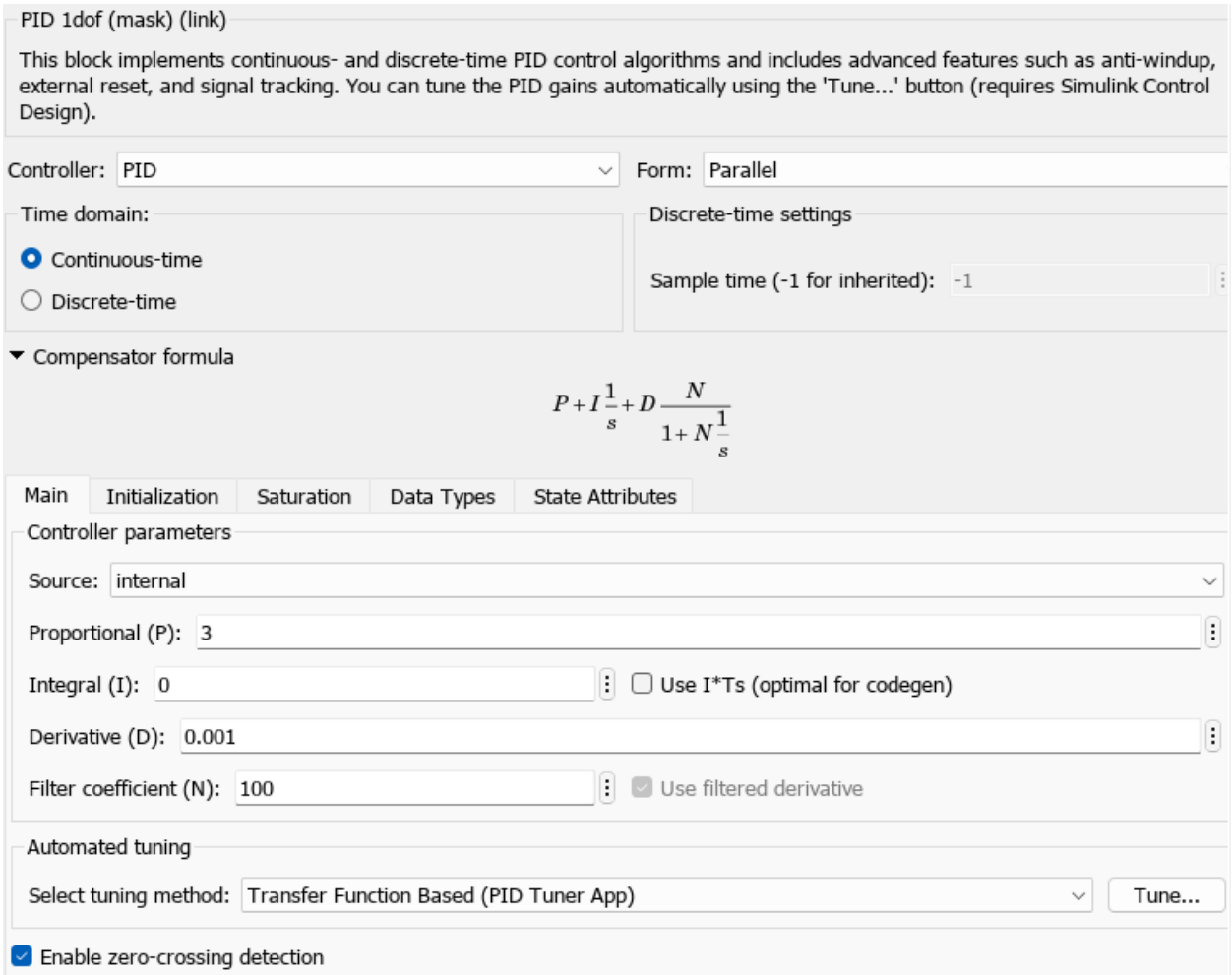


Kp=3

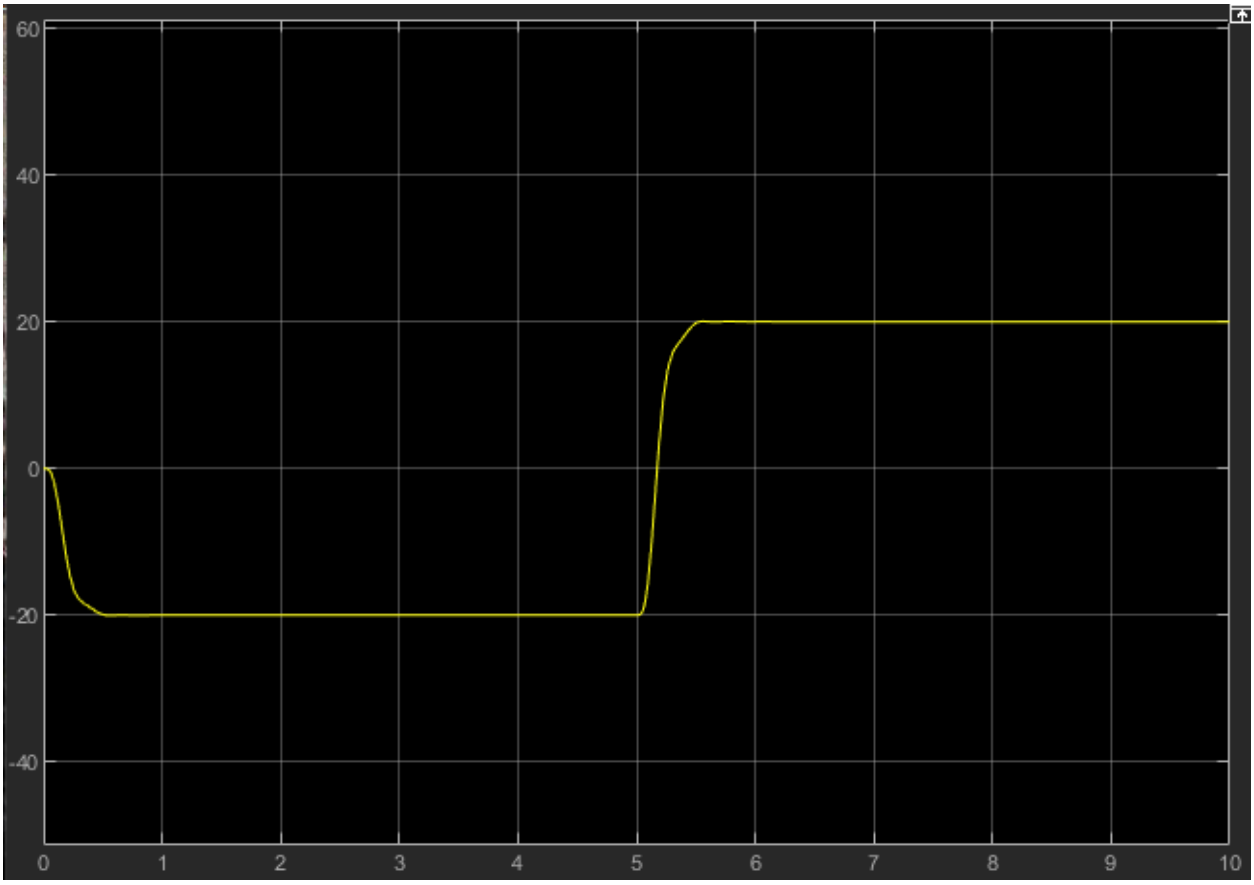
Ki=0

Kd=0

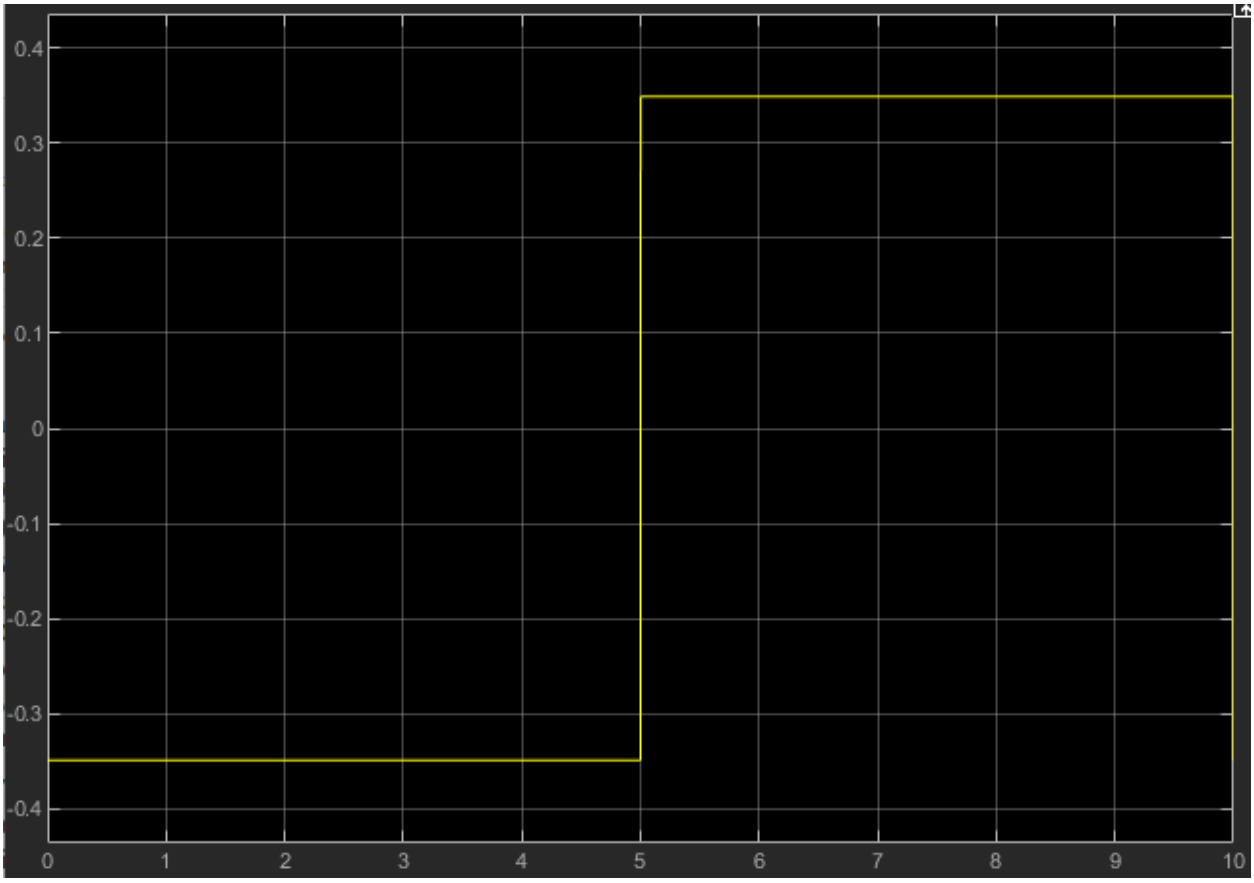
PD:


PID 1dof (mask) (link)
This block implements continuous- and discrete-time PID control algorithms and includes advanced features such as anti-windup, external reset, and signal tracking. You can tune the PID gains automatically using the 'Tune...' button (requires Simulink Control Design).
Controller: PID
Form: Parallel
Time domain:
Continuous-time
Discrete-time
Discrete-time settings
Sample time (-1 for inherited): -1
Compensator formula
$P + I\frac{1}{s} + D\frac{N}{1+N\frac{1}{s}}$
Main
Initialization
Saturation
Data Types
State Attributes
Controller parameters
Source: internal
Proportional (P): 3
Integral (I): 0
Use I*Ts (optimal for codegen)
Derivative (D): 0.001
Filter coefficient (N): 100
Use filtered derivative
Automated tuning
Select tuning method: Transfer Function Based (PID Tuner App)
Tune...
Enable zero-crossing detection

Theta+alpha vs. time



Voltage vs. time



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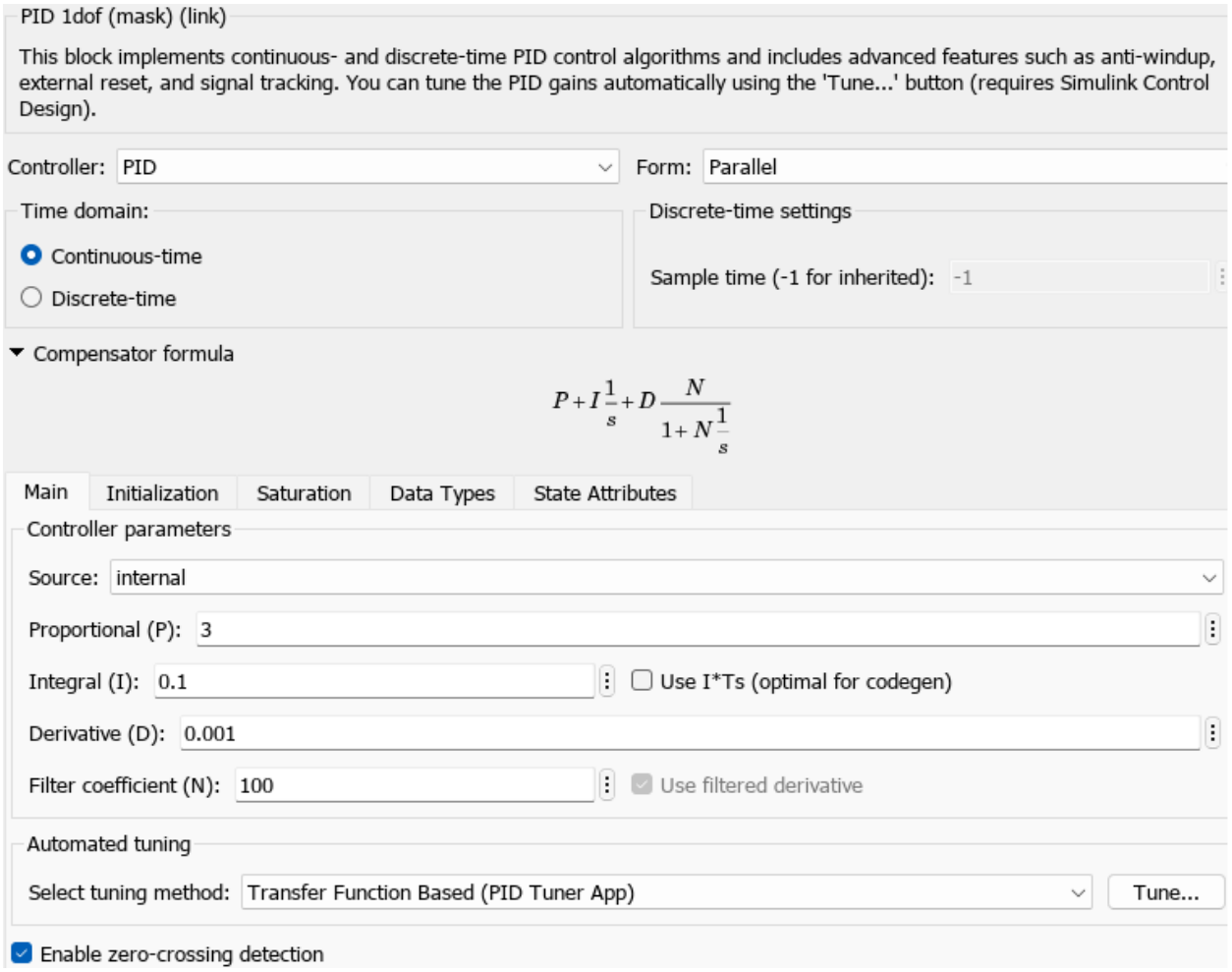
Ki=0

Kd=0.001

PID: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$P + I\frac{1}{s} + D\frac{N}{1 + N\frac{1}{s}}$
Main
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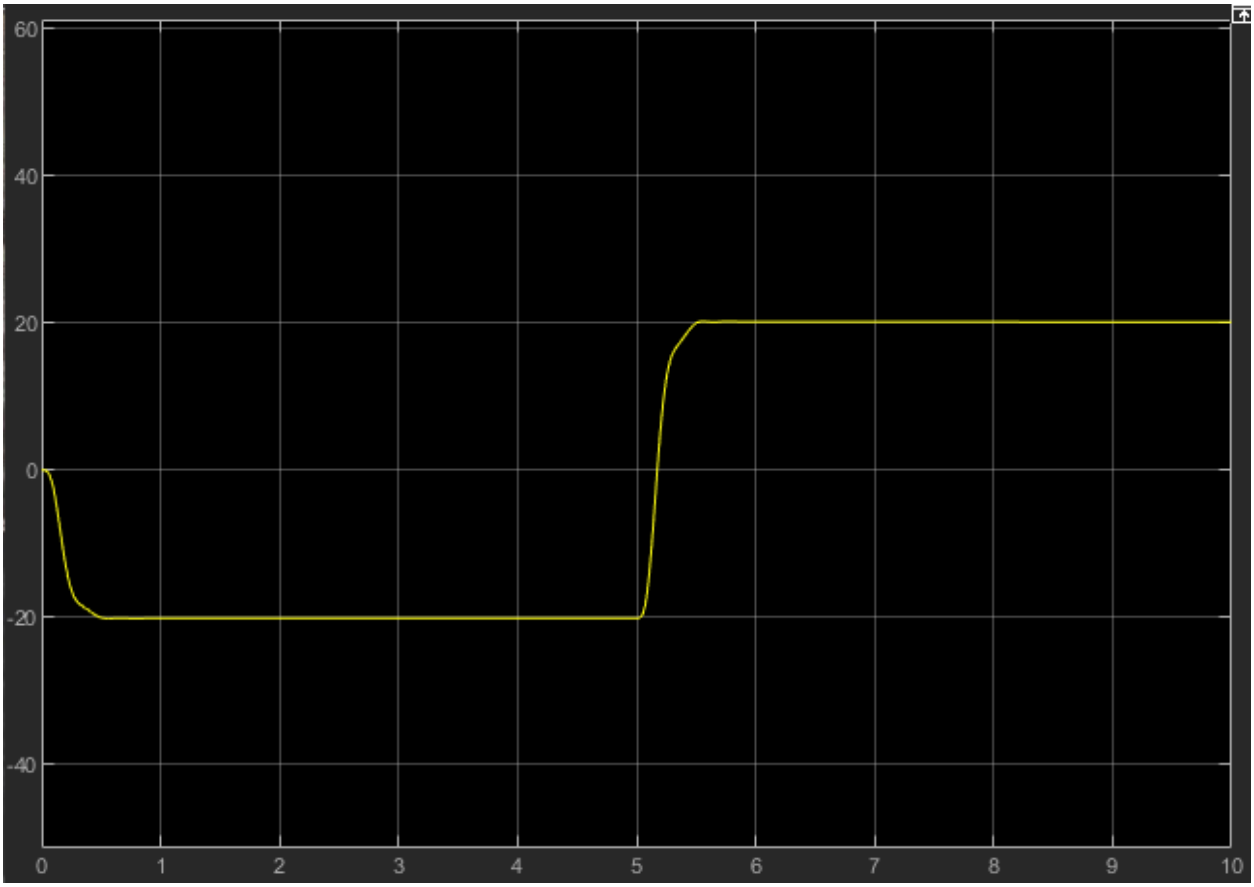
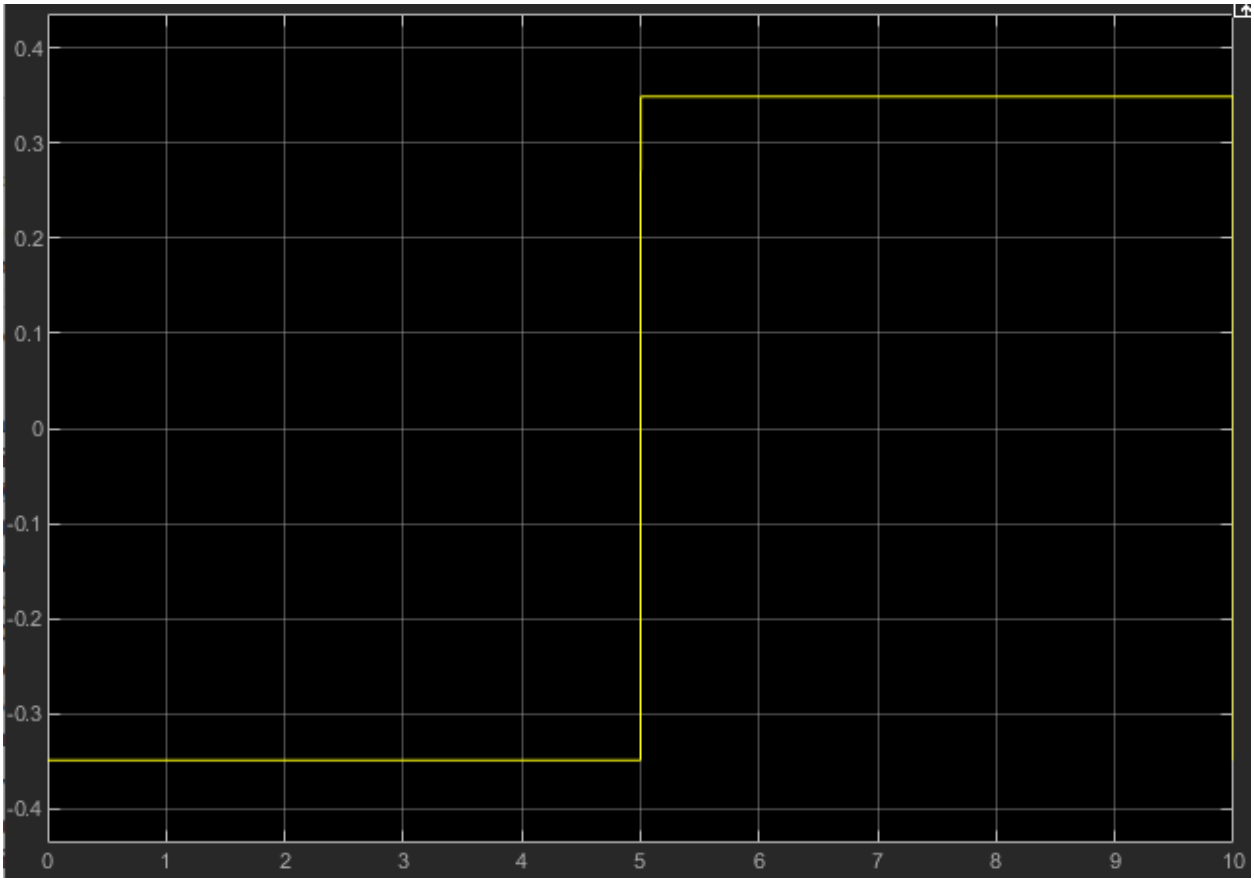
Theta+alpha vs. time



Voltage vs. time



Kp=3

Ki=0.1

Kd=0.001

All the design specifications are met.